

Earnings and wealth under four-year decisions

Operational diagnostic review of the native financing candidate

September 22, 2026 | source packet frozen after the 07:50 EDT search stop

Executive reading

The persistent-plus-iid earnings construction is operationally viable at the tested points: the full smoke passed, the direct four-year process uses 45 income states, and the expanded 160-node wealth grid reaches [-12, 3000] with zero assets at age 18. The evidence remains diagnostic. The longer search scored 7 of 8 proposals, stopped on a native timeout during proposal 006, and never completed final repeats or selection verification.

The candidate has four-year persistence 0.77614467, persistent innovation SD 0.43743545, and iid SD 0.16499061. These are direct four-year estimates from the saved data construction. Bootstrap evidence is weak for the iid component: its 199-draw bootstrap interval for the transitory variance runs from approximately zero to 0.06398. The point fit of three covariance moments to three parameters is mechanical and is not a proof of identification.

Across the three complete points, the beta=.985 smoke has the lowest loss (1040.447). The anchor is 1160.376 and search case005 is 1087.282. Case005 improves first-birth timing but worsens childlessness, occupied rooms, and ownership. It is a useful trade-off diagnostic, not an adopted calibration.

The fit trade-off at a glance

Moment	Target	Anchor	Beta 0.985	Search 005
First-birth age	25.98	28.21	28.11	26.42
Childlessness (%)	19.83	18.60	18.86	11.25
Mean rooms	5.56	6.45	6.43	6.94
Ownership 30-55 (%)	64.83	56.28	56.08	50.87

The earnings candidate works at the tested points. A defensible final calibration is still unfinished. Full target and parameter tables follow; no specification has been adopted.

What is established, and what is not

Each of the three reported points completed a normalized solution requiring six stationary solves. The starting point reproduced exactly across two runs; the beta 0.985 smoke point and search proposal 005 were each evaluated once. The saved contract keeps 13 target rows, with 12 scored moments plus a separately imposed completed-fertility normalization. Nine structural coordinates are free; the remaining rows are fixed, derived, or restricted.

The long search had 49 stationary solves started, 48 completed, and 1 incomplete. Including V5 smoke evidence, the honest count is 67 started, 66 completed, and 1 incomplete; the all-attempt count is 102 started, 98 completed, and 4 incomplete. The controller recorded a fatal contract error, but the observed cause is the native 3,100-second timeout; there is no evidence of a changed target or source fingerprint.

The income process follows the architecture precedent in Sommer (2016 JME): an AR(1) persistent component plus an iid component without a fixed permanent type. De Nardi (2004 ReStud) and Bick (2016 JEEA) support direct estimation at multi-year model frequencies, but their five-year and three-year objects are not drop-in four-year parameters. The saved literature review therefore supports the architecture and timing choice while leaving external parameter comparisons qualified.

What the earnings estimates measure

The PSID input measures gross reference-person plus spouse labor earnings. We sum four consecutive observed annual values, then remove block-start age and year effects from log block means. The selected annual-era sample (1984-1997 survey-year labels, ages 25-60) has 7,812 blocks for 3,932 persons, with 3,836 lag-one and 1,449 lag-two pairs. Complete positive-earnings histories are a selected sample; this is not a post-1997 estimate.

There is no fixed permanent type. The persistent component follows an AR(1), and the second component is iid. Four-year income is known when choices are made. Current income enters purchase eligibility, with accounting checked to avoid spending it twice. Neither direct estimation nor that accounting check resolves within-period timing.

Zero assets at entry age 18 and stationary initial earnings risk are external candidate assumptions: the estimation sample has no ages 18-24. The wealth grid is a numerical domain, not an economic saving cap. Its extension is not convergence evidence. Original figures retain crowded/clipped 45-state legends and wealth-axis compression; these inherited presentation limits are disclosed.

Target fit: Anchor

Loss = 1160.376138. Each row is read from the saved target-fit CSV; values are rounded for display directly from the saved records. The fertility normalization is separate from the scored objective.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Fertility normalization (unscored)	2.1	2.100014	1.44746e-05	-	-
Childlessness	0.1982788	0.1860016	-0.01227712	35,532.30	5.3557
Exactly one child	0.2136553	0.2479868	0.0343315	26,952.82	31.768
Mean first-birth age	25.97626	28.20612	2.229852	139.828	695.259
First births at age 30+	0.249278	0.3582245	0.1089465	13,866.07	164.581
Wealth / annual labor earnings	6.145861	6.443145	0.2972841	7.5951	0.671238
Annual bequests / wealth	0.0088	0.007231417	-0.001568583	5,165,289.26	12.709
Old-age wealth dispersion	3.515935	3.258016	-0.2579189	10.6164	0.706223
Mean occupied rooms	5.561097	6.447662	0.8865646	128.021	100.624
Ownership ages 30-55	0.648334	0.5628063	-0.08552769	2,339.36	17.1124
First-birth rooms response	0.7202463	1.338844	0.6185982	137.565	52.6412
Family rooms response	0.3470669	0.3172022	-0.0298647	280.528	0.250203
Recent-parent ownership difference	0.1628955	0.1089627	-0.05393281	27,055.82	78.6986

Interpretation: the table is descriptive evidence under an unchanged target contract. A lower scalar loss at one point does not establish a local optimum, identification, or adoption.

Target fit: beta=.985 smoke

Loss = 1040.446863. Each row is read from the saved target-fit CSV; values are rounded for display directly from the saved records. The fertility normalization is separate from the scored objective.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Fertility normalization (unscored)	2.1	2.100011	1.084827e-05	-	-
Childlessness	0.1982788	0.1886469	-0.009631823	35,532.30	3.2964
Exactly one child	0.2136553	0.2461067	0.03245141	26,952.82	28.3838
Mean first-birth age	25.97626	28.11245	2.13619	139.828	638.078
First births at age 30+	0.249278	0.3532941	0.1040161	13,866.07	150.022
Wealth / annual labor earnings	6.145861	5.95471	-0.1911514	7.5951	0.277516
Annual bequests / wealth	0.0088	0.00723873	-0.00156127	5,165,289.26	12.5907
Old-age wealth dispersion	3.515935	3.36183	-0.1541051	10.6164	0.252121
Mean occupied rooms	5.561097	6.432706	0.8716088	128.021	97.2576
Ownership ages 30-55	0.648334	0.5607543	-0.08757978	2,339.36	17.9434
First-birth rooms response	0.7202463	1.266081	0.5458352	137.565	40.9857
Family rooms response	0.3470669	0.308851	-0.03821596	280.528	0.4097
Recent-parent ownership difference	0.1628955	0.1195004	-0.04339508	27,055.82	50.9497

Interpretation: the table is descriptive evidence under an unchanged target contract. A lower scalar loss at one point does not establish a local optimum, identification, or adoption.

Target fit: Search case005

Loss = 1087.281615. Each row is read from the saved target-fit CSV; values are rounded for display directly from the saved records. The fertility normalization is separate from the scored objective.

Target moment	Target	Model	Gap	Actual weight	Loss contribution
Fertility normalization (unscored)	2.1	2.100141	0.0001405396	-	-
Childlessness	0.1982788	0.1124747	-0.085804	35,532.30	261.6
Exactly one child	0.2136553	0.2954333	0.08177795	26,952.82	180.251
Mean first-birth age	25.97626	26.42369	0.4474247	139.828	27.992
First births at age 30+	0.249278	0.2509093	0.001631269	13,866.07	0.0368982
Wealth / annual labor earnings	6.145861	6.021592	-0.1242692	7.5951	0.11729
Annual bequests / wealth	0.0088	0.004362503	-0.004437497	5,165,289.26	101.712
Old-age wealth dispersion	3.515935	4.43515	0.9192149	10.6164	8.97036
Mean occupied rooms	5.561097	6.936291	1.375193	128.021	242.107
Ownership ages 30-55	0.648334	0.5087247	-0.1396094	2,339.36	45.596
First-birth rooms response	0.7202463	1.021935	0.3016885	137.565	12.5206
Family rooms response	0.3470669	0.2867633	-0.06030367	280.528	1.02015
Recent-parent ownership difference	0.1628955	0.0757739	-0.08712161	27,055.82	205.358

Interpretation: the table is descriptive evidence under an unchanged target contract. A lower scalar loss at one point does not establish a local optimum, identification, or adoption.

All 17 parameters: estimates, bounds, and restrictions

The parameter comparison below distinguishes actual bounds from fixed or derived objects. Near-bound flags are recalculated from the actual frozen plan: within 1% of each raw bound width. A / B / S denote the anchor, beta 0.985 test and search proposal. Annual beta is bounded above by 0.99, and the parenthood floor by 2.3. These flags do not establish identification. At search case005, beta_annual is at its actual upper bound and theta0 is at zero, so theta1 is inactive in the implemented bequest utility.

Parameter	Anchor	Beta 0.985	Search 005	Actual bounds / restriction	Near: A / B / S
Annual discount factor	0.99	0.985	0.99	[0.94, 0.99]	Y / N / Y
First-birth shock scale	0.3377342	0.3377342	0.350502	[0.02, 50]	Y / Y / Y
Later-birth shock scale	0.3978565	0.3978565	0.6768031	[0.02, 50]	Y / Y / N
Owner service multiplier	1.049653	1.049653	0.9233887	[0.1, 5]	N / N / N
Supply intercept	8.1121	8.1121	10.51613	[0.2, 80]	N / N / N
Bequest scale	0.08105103	0.08105103	0	[0, 8]	N / N / Y
Bequest wealth shift	0.08520357	0.08520357	0.1196294	[0.02, 16]	Y / Y / Y
First-birth utility cost	0.2657648	0.2657648	0.1235503	[0, 8]	N / N / N
Parenthood housing floor	2.3	2.3	1.914603	[0.1, 2.3]	Y / Y / N

All 17 parameters continued

Parameter	Anchor	Beta 0.985	Search 005	Actual bounds / restriction	Near: A / B / S
Extra floor per child	0	0	0	zero restriction	-
Fertility utility intercept	0.1910551	0.1924098	0.07848538	normalized to 2.1	-
Payroll tax rate	0.179	0.179	0.179	externally fixed	-
Four-year pension	2.046361	2.046361	2.046361	budget derived	-
Supply elasticity	0.63	0.63	0.63	externally fixed	-
Tenure shock scale	0.005	0.005	0.005	externally fixed	-
Consumption exponent	0.733	0.733	0.733	externally fixed	-
Utility curvature	2	2	2	externally fixed	-

Fixed and derived rows are included to make the full contract visible. The reported theta1 value is not identified when theta0=0; this is a structural inactivity statement, not a claim that the optimizer found a unique value.

Units and economic interpretation

At search case005, annual beta 0.99 implies a four-year discount factor of 0.96059601. The parenthood housing floor is 1.9146 rooms when children are resident; the additional floor per child is zero. H0 is the housing supply intercept, and chi multiplies owner housing services.

The bequest scale theta0 is zero at this point, so the bequest wealth shift theta1 has no effect on utility here. Positive accidental bequests may still occur. The fertility and tenure kappa parameters are taste-shock scales: their numerical sizes depend on the utility normalization and have no directly portable literature benchmark.

Numerical limits and next steps

The search result should be treated as a partial operational review. Seven valid proposals are insufficient to characterize a frontier, and the best valid proposal was evaluated once. The completed smoke point at $\beta = .985$ is outside the frozen search candidate pool, so comparing its loss to case005 is informative but not a selection rule.

Earlier attempts exposed three implementation problems: transactions clipped outside grid support, loss of a sale option at the old wealth-grid ceiling, and reported consumption/housing floors that differed from the optimizer allocation. Separately reviewed corrections passed targeted checks; older evidence is preserved. No economic target or numerical gate was relaxed.

The principal numerical limits are the unverified continuation/interpolation behavior at tiny-mass states, wealth-grid density and domain sensitivity, and the absence of exact repetitions for case005. The all-17 graph packet remains the stable diagnostic set. No evidence supports dropping or demoting any target, and no unreachable-target conclusion is warranted.

Recommended next sequence: preserve the direct four-year construction; repair and stub-test the timeout/failure path before another long run; declare the selection pool and any changed duration in a new frozen plan; resolve the continuation and grid diagnostics; then repeat a declared candidate exactly before a further focused calibration under the same targets. Inspect the full target and parameter tables before paper adoption.

These are empirical estimates, with material uncertainty. The 199-draw person bootstrap gives persistence 0.7211-0.8348, persistent variance 0.4302-0.5249, and iid variance approximately 0-0.0640 (95% percentile intervals). The iid component is weakly estimated. Precision, the entrant distribution and within-period timing still need to be defended before paper adoption.

References and provenance

Sommer, Kamila (2016). "Fertility Choice in a Life Cycle Model with Idiosyncratic Uninsurable Earnings Risk." Journal of Monetary Economics 83, 27-38. Author PDF: <https://www.kamilasommer.net/Fertility.pdf>

De Nardi, Mariacristina (2004). "Wealth Inequality and Intergenerational Links." Review of Economic Studies 71(3), 743-768. Primary PDF: <https://users.nber.org/~denardim/research/denardi.pdf>

Bick, Alexander (2016). "The Quantitative Role of Child Care for Female Labor Force Participation and Fertility." Journal of the European Economic Association 14(3), 639-668. Article record: <https://onlinelibrary.wiley.com/doi/abs/10.1111/jeea.12143>

Saved project reviews: literature/annual_sources_review.md; literature/multiyear_review.md; literature/parameter_validation_rubric.md; terminal_review.md. These sources distinguish architecture precedent from directly estimated four-year parameters.

Report provenance

Generated from /Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26/output/model/native_financing_diagnostic_20260919/specification_followup/earnings_wealth_v1. Source data are read at build time. Graphs are copied by reference into the PDF from search_readout/standard_diagnostics and are not edited or regenerated.

Frozen run fingerprints (SHA-256)

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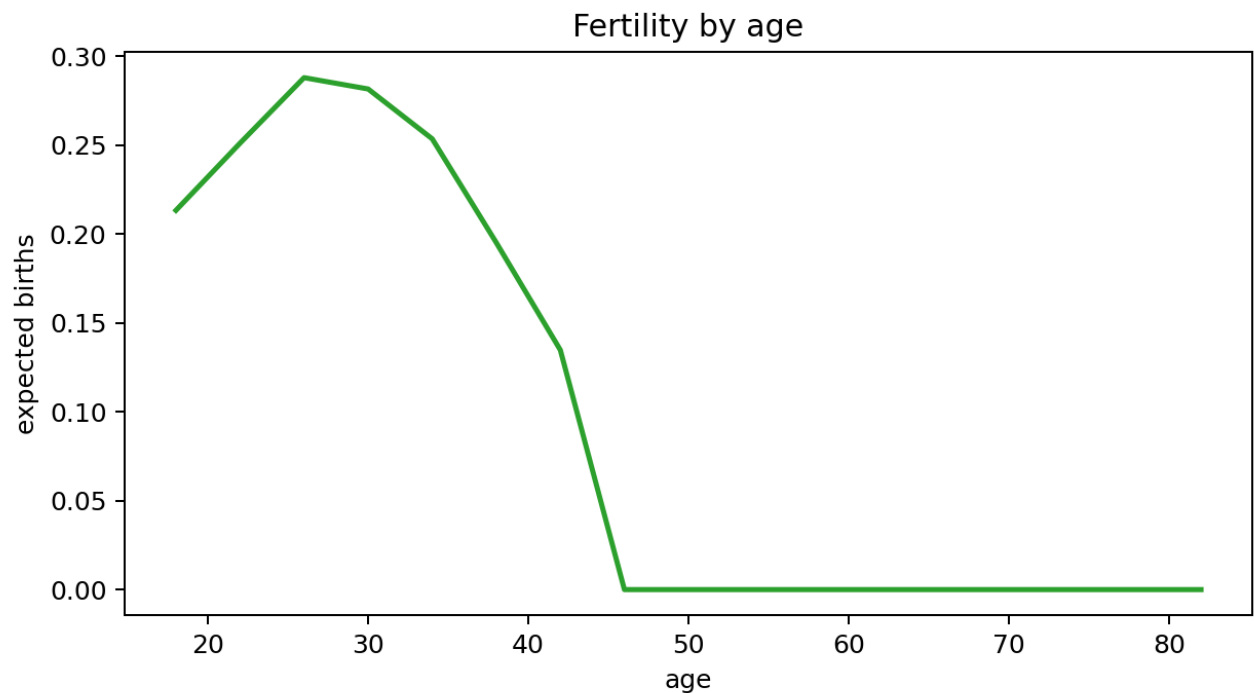
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Economic source manifest: 612ea34da2f462cb50dcf6881dd3005d8495f827c833525436b150500ebf11cf

Exact anchor repetition checks cover prices, value functions, population arrays and moments. No such completed final comparison exists for the improved smoke point or search case005. Full generated-code and target provenance are retained in the source packet.

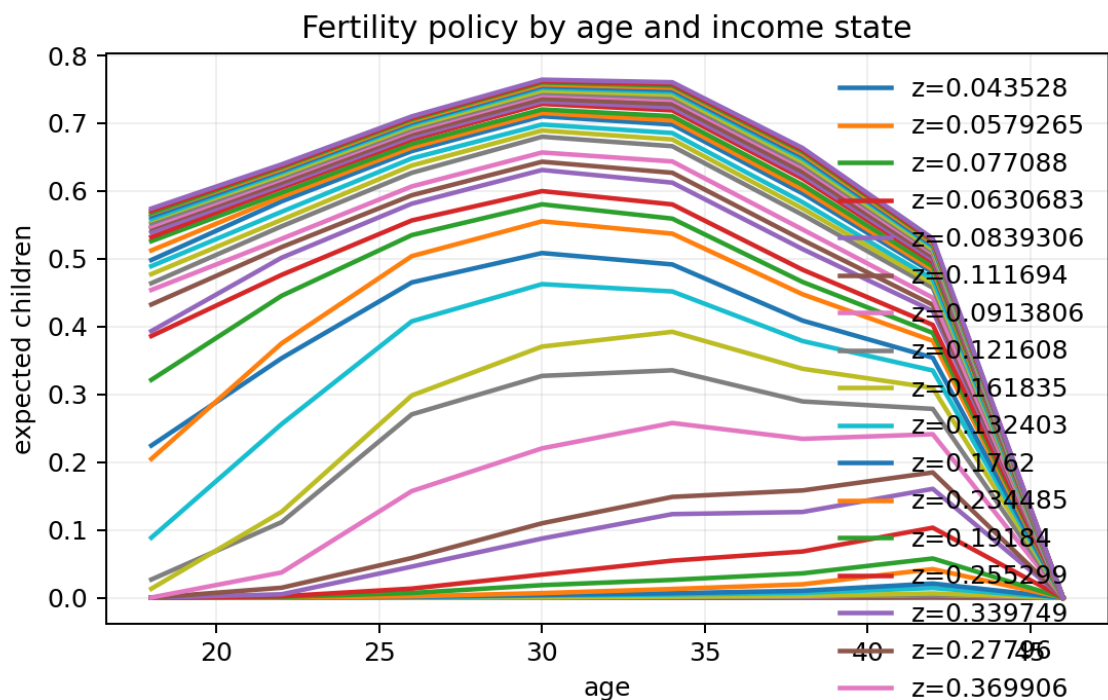
Appendix: original standard diagnostic plots (1-2 of 17)

fertility by age



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

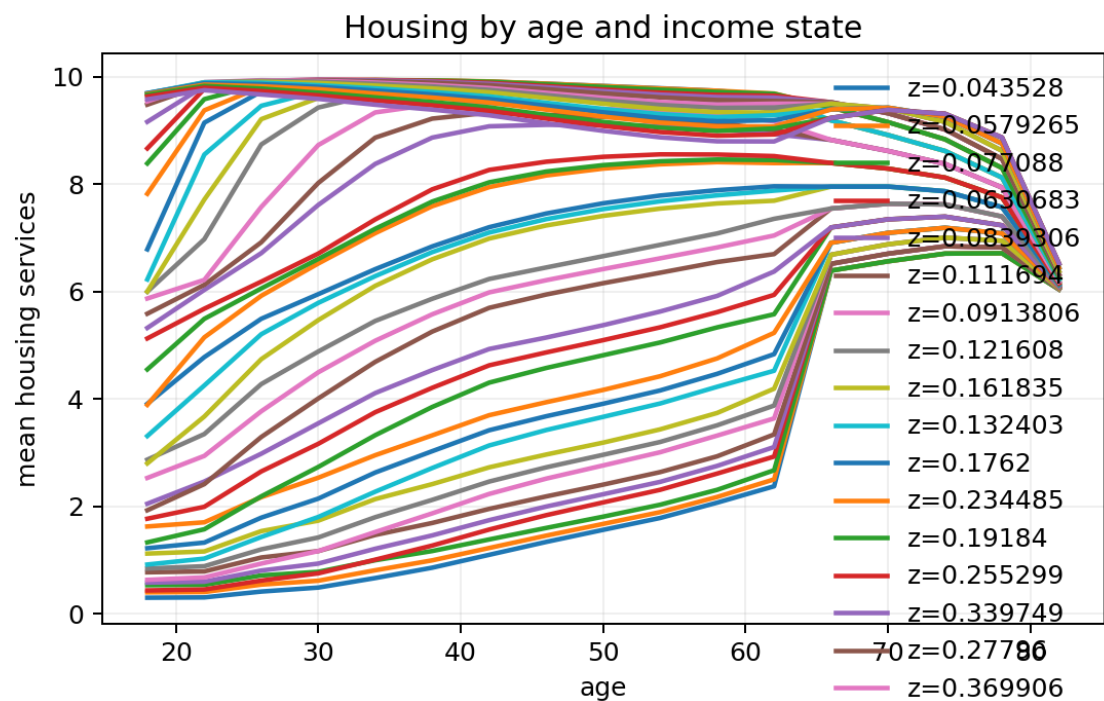
fertility policy by age income state



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

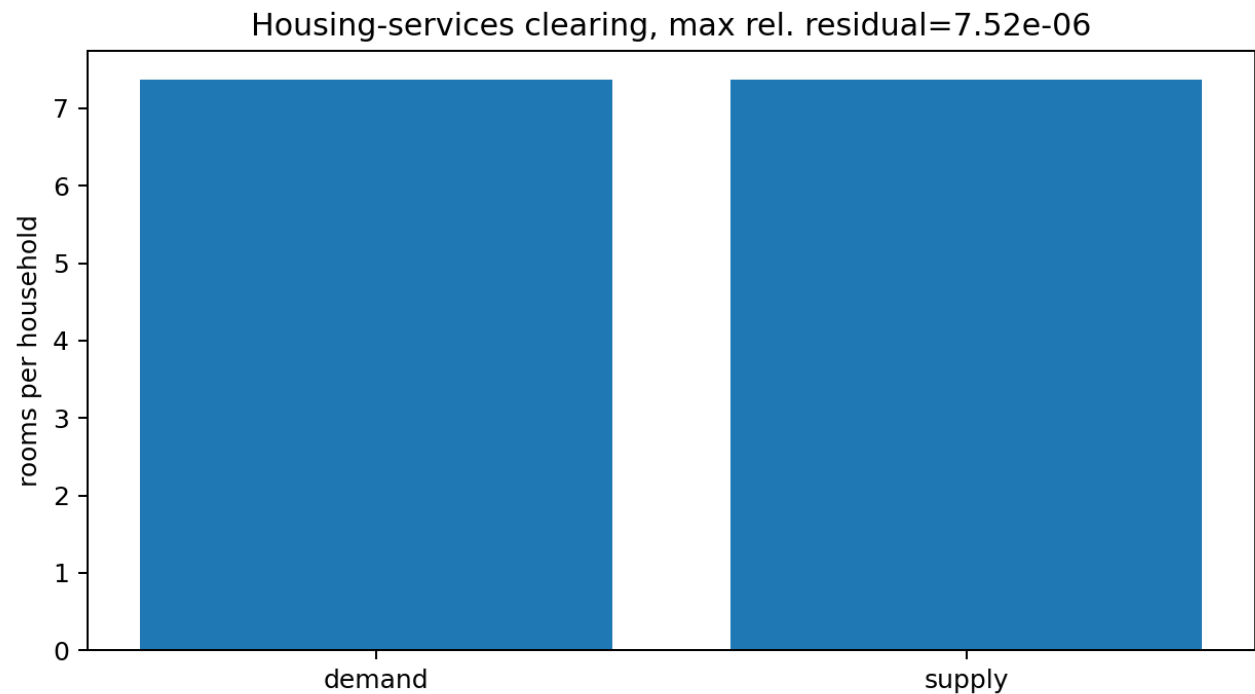
Appendix: original standard diagnostic plots (3-4 of 17)

housing by age income state



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

housing market



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

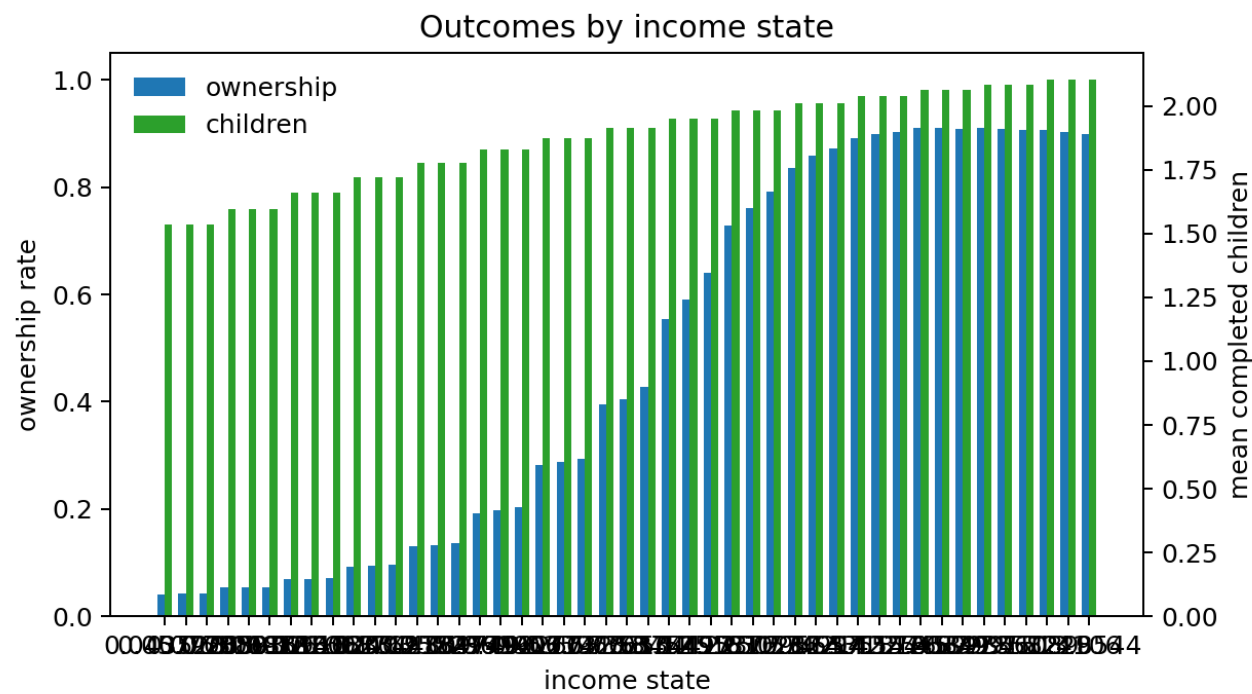
Appendix: original standard diagnostic plots (5-6 of 17)

housing prices



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

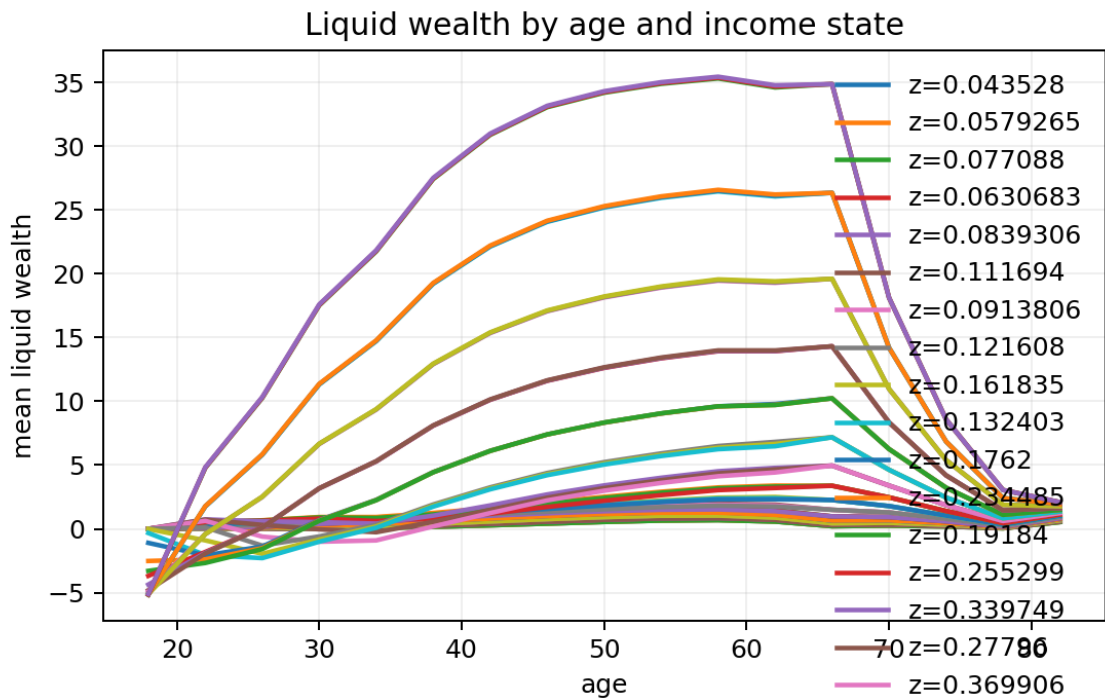
income state outcomes



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

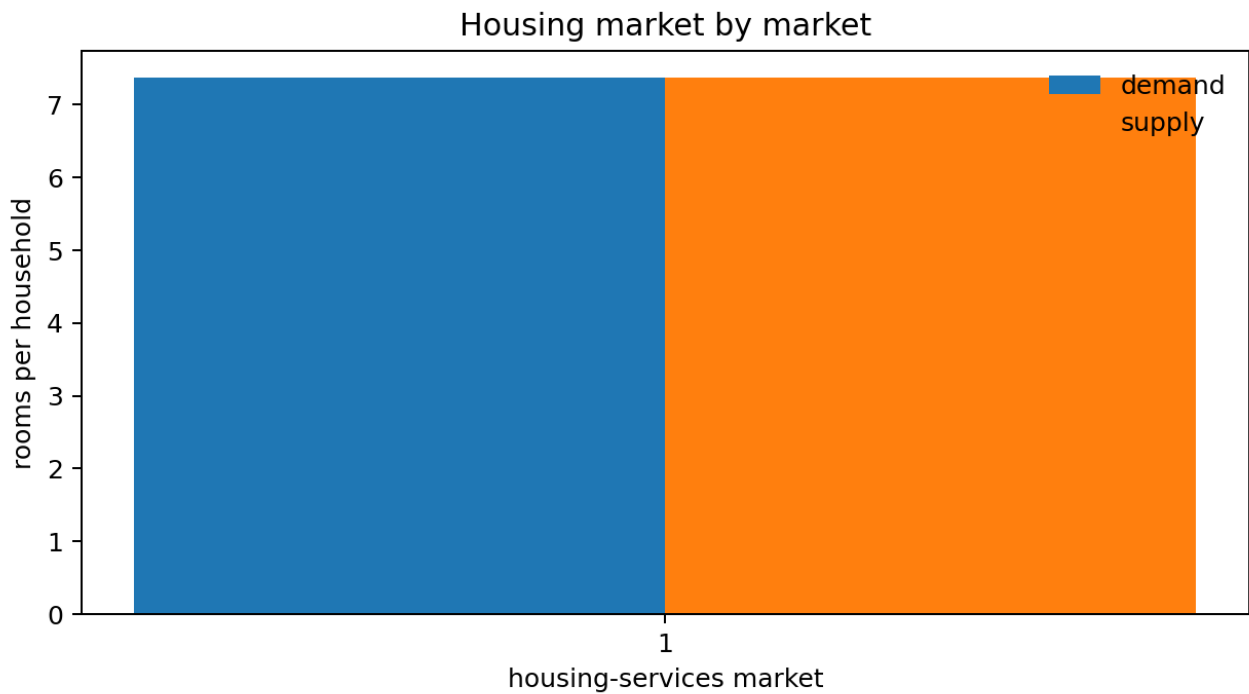
Appendix: original standard diagnostic plots (7-8 of 17)

liquid wealth by age income state



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

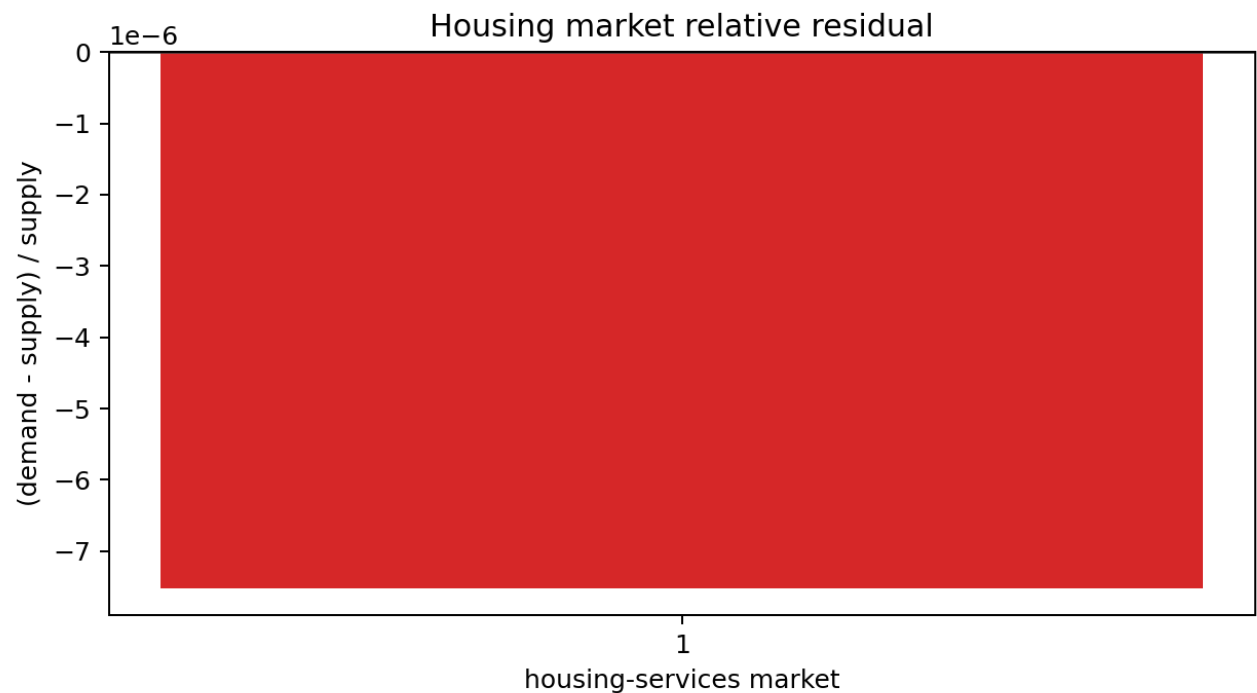
market clearing by market



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

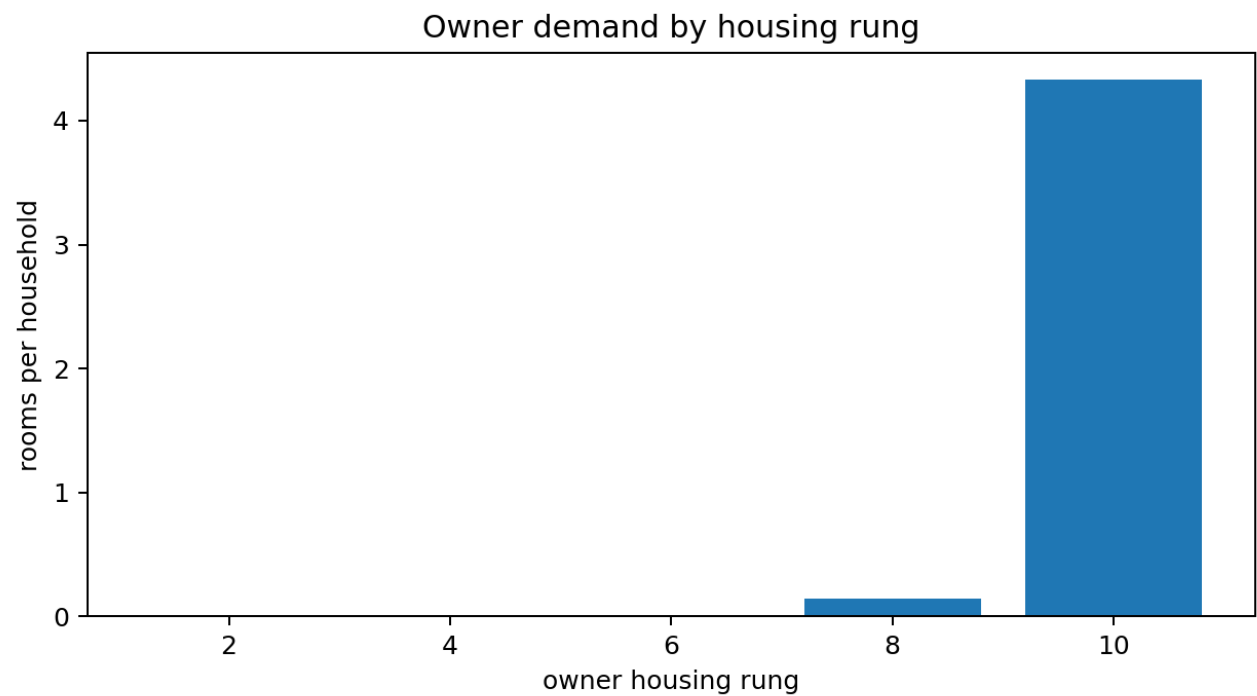
Appendix: original standard diagnostic plots (9-10 of 17)

market clearing residuals



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

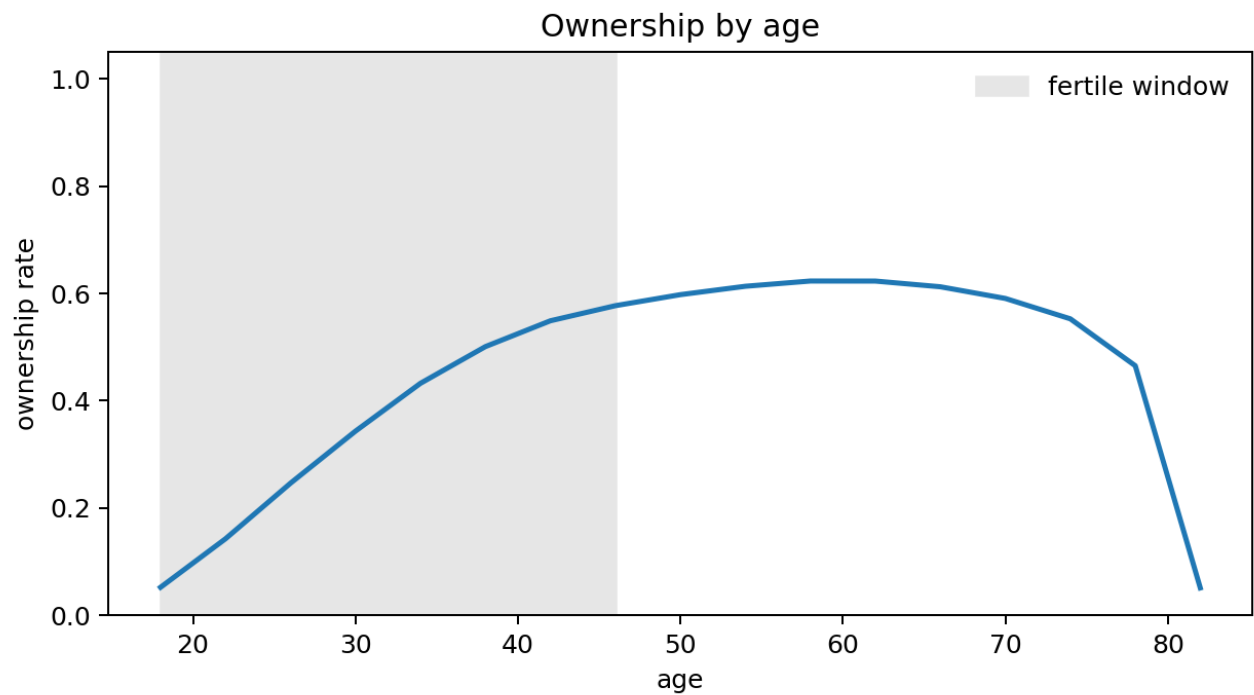
owner rungs



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

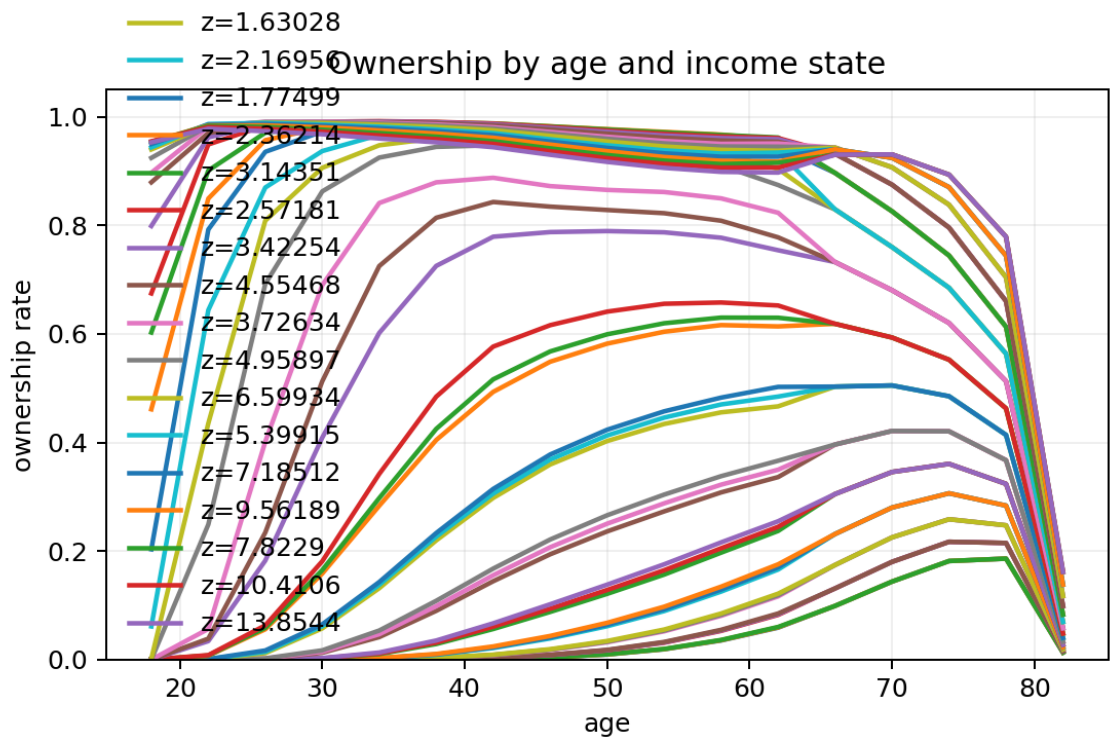
Appendix: original standard diagnostic plots (11-12 of 17)

ownership by age



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

ownership by age income state

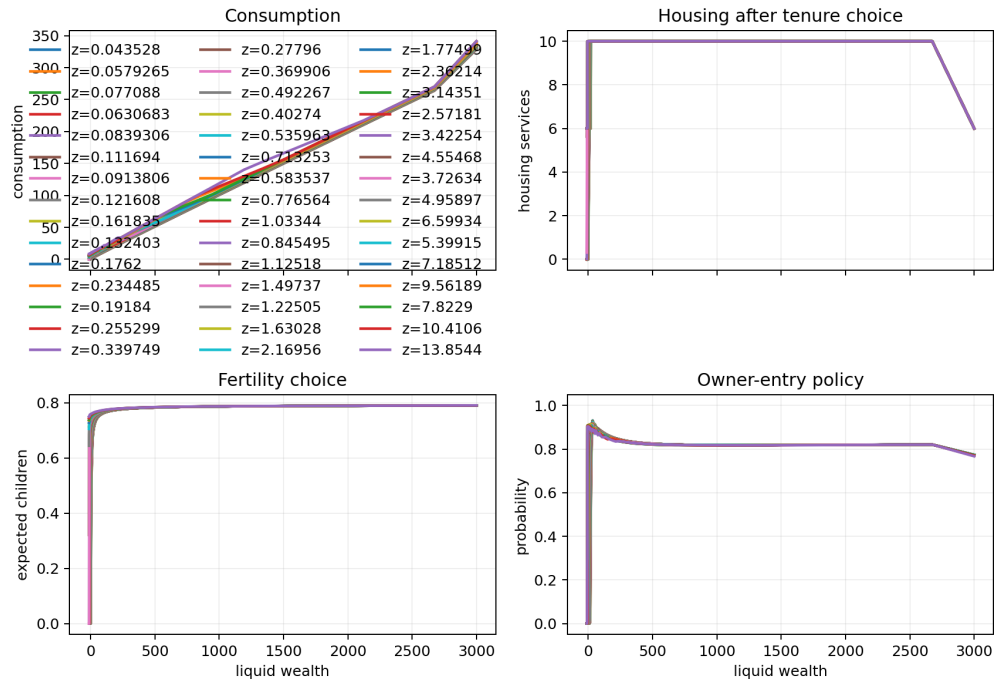


Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

Appendix: original standard diagnostic plots (13-14 of 17)

policy childless renter age30

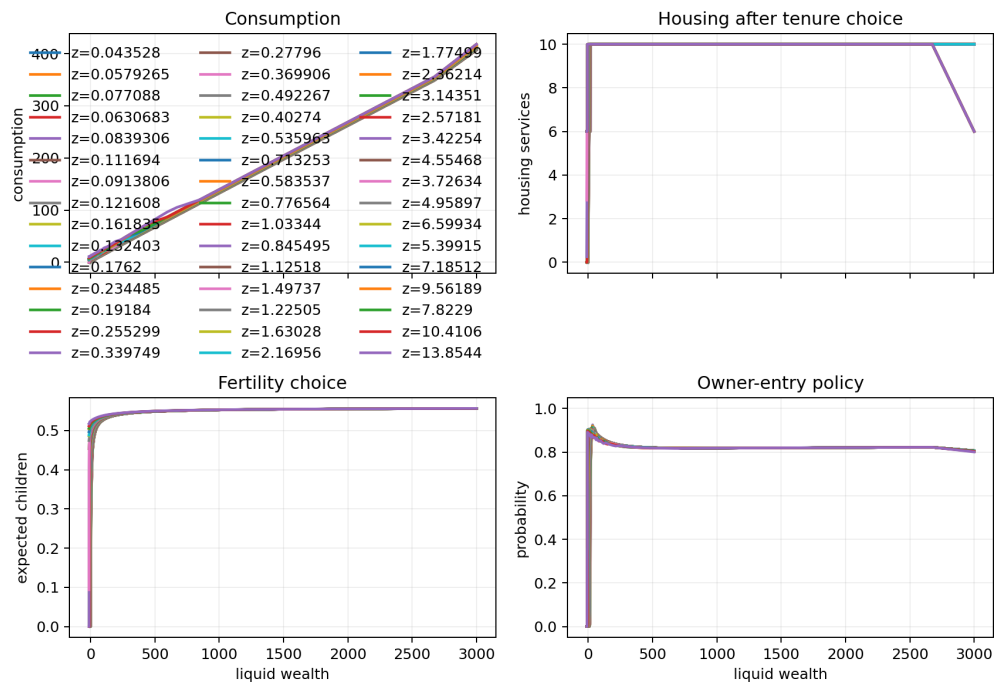
Childless renter policies at age 30



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

policy childless renter age42

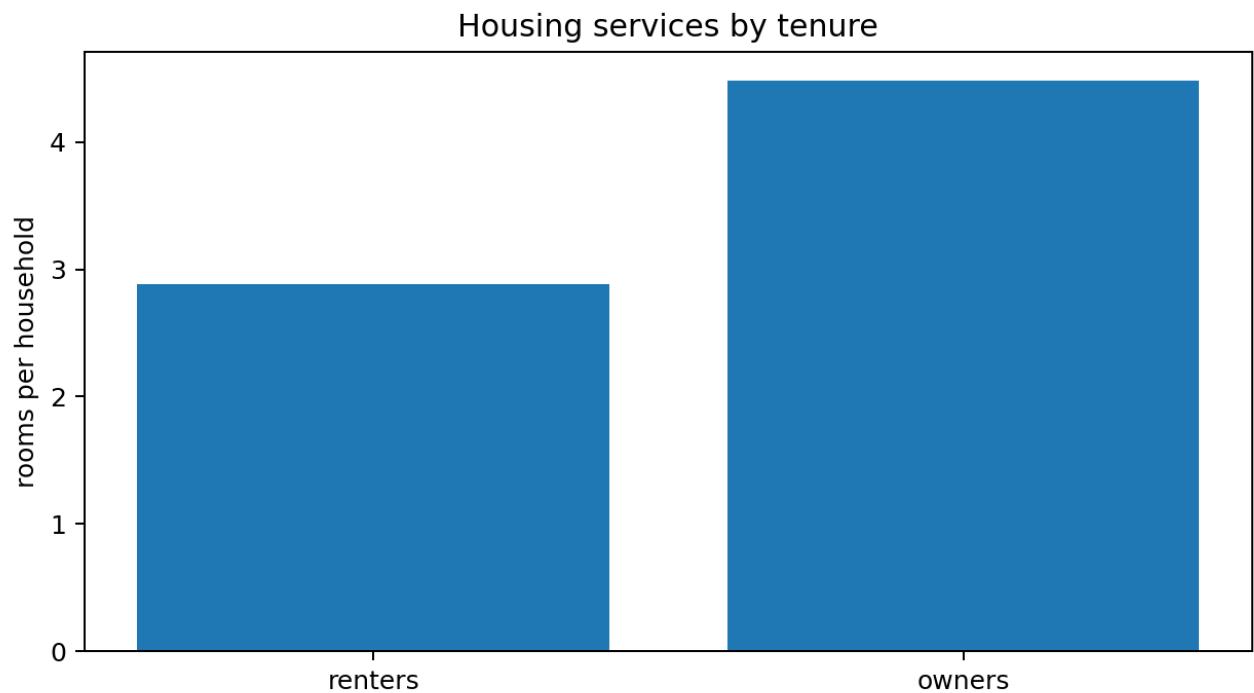
Childless renter policies at age 42



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

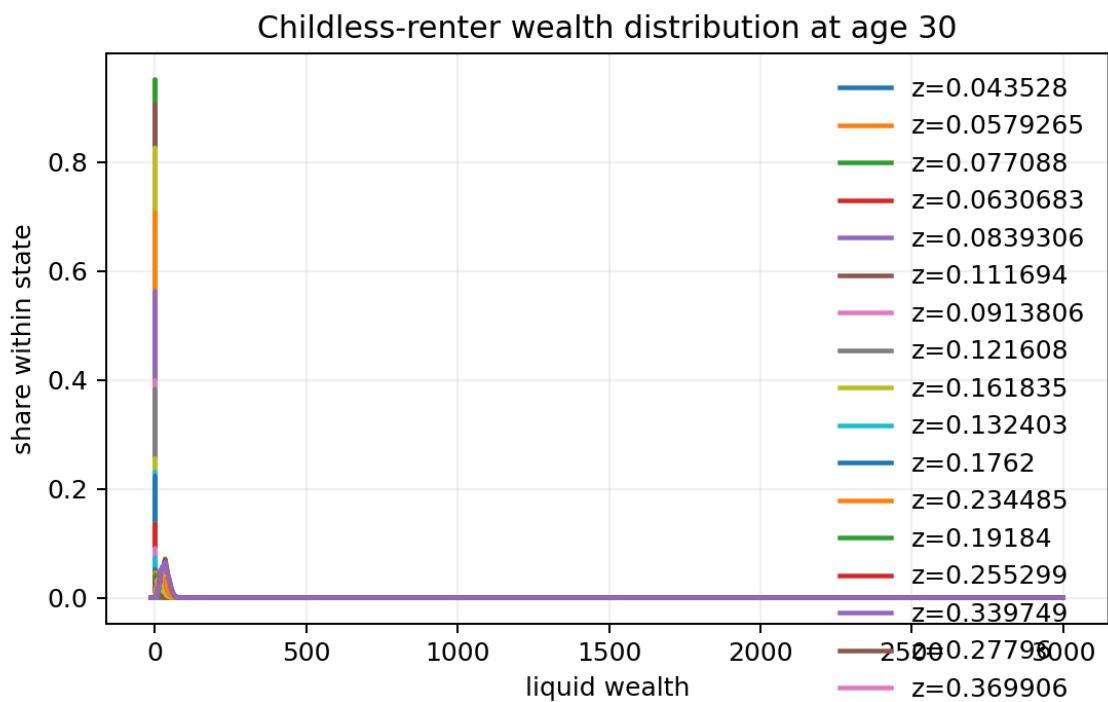
Appendix: original standard diagnostic plots (15-16 of 17)

tenure services



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

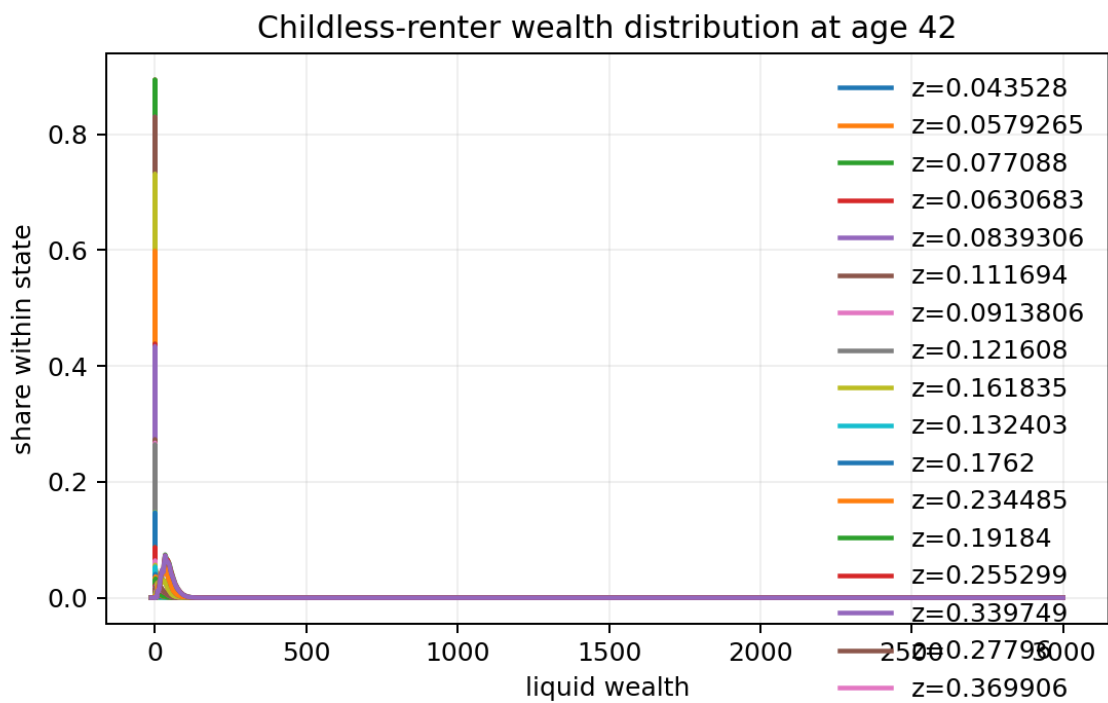
wealth dist childless renter age30



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.

Appendix: original standard diagnostic plots (17-17 of 17)

wealth dist childless renter age42



Original case005 graph retained without redesign. Read wealth-axis compression and crowded/clipped 45-state legends with care.